calculating the mean and standard deviation requires the time consuming operations of addition and multiplication. The strategy of this algorithm is to use these slow operations only on the few numbers in the histogram, not the many samples in the signal. This makes the algorithm much faster than the previously described methods. Think a factor of ten for very long signals with the calculations being performed on a general purpose computer.

The notion that the acquired signal is a noisy version of the underlying process is very important; so important that some of the concepts are given different names. The *histogram* is what is formed from an acquired signal. The corresponding curve for the underlying process is called the **probability mass function (pmf)**. A histogram is always calculated using a finite number of samples, while the pmf is what *would be* obtained with an infinite number of samples. The pmf can be estimated (inferred) from the histogram, or it may be deduced by some mathematical technique, such as in the coin flipping example.

Figure 2-5 shows an example pmf, and one of the possible histograms that could be associated with it. The key to understanding these concepts rests in the units of the vertical axis. As previously described, the vertical axis of the histogram is the *number of times* that a particular value occurs in the signal. The vertical axis of the pmf contains similar information, except expressed on a *fractional basis*. In other words, each value in the histogram is divided by the total number of samples to approximate the pmf. This means that each value in the pmf must be between zero and one, and that the sum of all of the values in the pmf will be equal to one.

The pmf is important because it describes the *probability* that a certain value will be generated. For example, imagine a signal with the pmf of Fig. 2-5b, such as previously shown in Fig. 2-4a. What is the probability that a sample taken from this signal will have a value of 120? Figure 2-5b provides the answer, 0.03, or about 1 chance in 34. What is the probability that a randomly chosen sample will have a value greater than 150? Adding up the values in the pmf for: 151, 152, 153,···, 255, provides the answer, 0.0122, or about 1 chance in 82. Thus, the signal would be expected to have a value exceeding 150 on an average of every 82 points. What is the probability that any one sample will be between 0 and 255? Summing all of the values in the histogram produces the probability of 1.00, a certainty that this will occur.

The histogram and pmf can only be used with discrete data, such as a digitized signal residing in a computer. A similar concept applies to continuous signals, such as voltages appearing in analog electronics. The **probability density function (pdf)**, also called the **probability distribution function**, is to continuous signals what the probability mass function is to discrete signals. For example, imagine an analog signal passing through an analog-to-digital converter, resulting in the digitized signal of Fig. 2-4a. For simplicity, we will assume that voltages between 0 and 255 millivolts become digitized into digital numbers between 0 and 255. The pmf of this digital